

# Revision to Proposition 6.4

*The density approximation at display (55)*

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## What was missing

The density statement after Lemma 6.3 applies to the algebra generated by digit-cylinder indicators, continuous functions of the real window coordinates, and characters in all torus window coordinates. In the initial version, display (55) then selected approximants from the smaller monomial class (32), without writing the reduction from the larger algebra to that class. This matters because Proposition 4.2 is stated only for finite linear combinations of the monomials in (32).

## What I added in v5

I inserted the reduction explicitly. For a fixed radius  $R$ , the revised proof follows the approximation

$$B^{(R)} \rightarrow G \rightarrow G_M \rightarrow G_M I_E \rightarrow P_{R,M}$$

- The projection from radius  $R+M$  to radius  $R$  places the fixed-radius function and the larger-radius approximants on the same window space.
- $G$  is chosen from the full dense algebra, including continuous dependence on the real coordinates and characters throughout the window.
- Each real coordinate is replaced by its first  $M$  future digits. Continued-fraction contraction gives an error of order the inverse square of the  $M$ -th Fibonacci number, with  $R$  fixed.
- I then restrict to the event on which the digits in the required finite word are at most  $K$ . Lemma 3.1(ii) bounds the discarded measure by  $O(M/K)$ , with  $R$  fixed.
- On each remaining digit cylinder, identity (31) reduces every torus character to a character in the central pair at  $j-1$  and  $j$ .
- For fixed  $M$  and  $K$ , only finitely many words and Fourier modes remain. The truncated function is therefore exactly a finite linear combination of monomials (32), denoted  $P(R,M)$ .

## Why the radius grows with $M$

Approximating the real coordinate at  $j+R$  by  $M$  future digits requires digits through  $j+R+M$ . The corresponding monomial must therefore be allowed to use a word of radius  $R+M$ . This is the reason display (55) uses radius  $R+M$ .

## Use of Proposition 4.2

The corrected display (55) measures the approximation error on the radius  $R+M$  window space. Lemma 6.3 transfers the squared error from the stationary window law to the actual windows. Since each approximant has fixed finite radius, digit support, and Fourier support before  $n$  tends to infinity, Proposition 4.2 applies directly to the variance calculation in (57). No additional theorem is used in this step. The relevant passage is on v5 PDF pp. 15-16; in the TeX source, the bridge is at lines 1307-1408 and the Proposition 4.2 step is at lines 1419-1436.